

PAM-Glider Noise Profiler

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Background/Justification

- Gliders are remarkably quiet, but must contend with self-noise¹, flow noise^{2,3} and external sources: wind⁴, human activity⁵.
- Noise may limit our ability to acoustically detect marine mammals by lowering signal-to-noise ratios (SNR), **reducing probabilities of detection**³.
- It is helpful to first identify the **conditions** which yield elevated noise levels:
 - What **variables** (EX. glider mode, depth, location, etc.) affect noise levels?
 - Are these effects **broadband** (affecting all frequencies) or **narrowband** (affecting only select frequency bands)?

Goals

- Create a tool which outputs an **automated slideshow-based report** summarizing how noise levels are linked to input variables of interest.
- Allow the user to **easily visualize** which conditions contribute to elevated noise levels, and which frequency bands are most affected.

Methods: Data

DATA: Seaglider SG607

(one of eight deployments)

Noise metrics

- Hybrid-millidecade (HMD) sound levels
- Sound pressure levels (SPL) (Broadband)

Instrument and environmental variables

- Glider mode
- Glider depth
- Temperature, salinity, seawater density, sound velocity (m/s)
- These are each **binned** prior to further analysis (except glider mode)

Coding master,
Julia Engdahl



Methods: Workflow

1. **INPUTS:** Predictors (Depth, mode, env. data) & responses (HMD levels, SPL)
2. **Build timetable:** Align glider mode, depth, and local sound speed to each HMD timestamp.
3. **Subset data:** HMD dataset is split into subsets along each predictor:
 - a. One overarching subset per glider mode
 - i. One subset per glider depth bin (100 m)
 - ii. One subset per local sound speed bin (5 m/s)
 - iii. Etc. for all environmental variables
4. **Plot each subset:** For each predictor, generate HMD plot with curves color-coded by bin.
5. **Output:** Compile all plots into an automated report (PPT)
Package: python_pptx

Final Product: Summary Slide Deck (Abbreviated)

sg607 PAM Noise Analysis

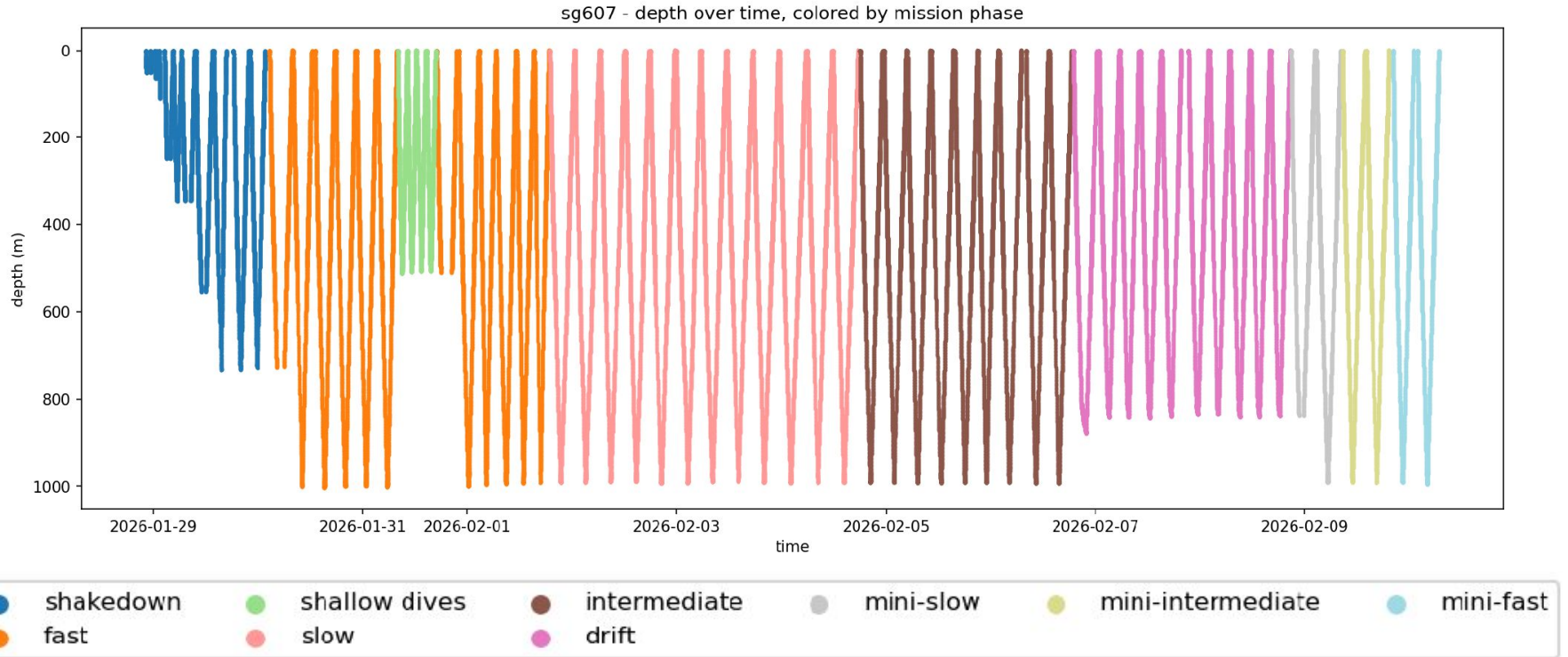
9 mission phases analyzed | QC
applied

Summary

- 124,836 science rows across 9 mission phases: shakedown, fast, shallow dives, slow, intermediate, drift, mini-slow, mini-intermediate, mini-fast
- Overall mean broadband SPL: 93.9 dB (min 78.4, max 136.9)
- Variables binned: temperature, salinity, density, soundVelocity, depth, latitude, longitude
- QC status: QC applied
- depth_mask_flag: 3,518 of 126,907 rows flagged (2.8%) - dropped from this analysis
- Strongest overall correlation with broadband SPL: latitude ($r=0.13$, $n=123,389$)
- Strongest per-mode correlation: depth during "shallow dives" ($r=0.26$, $n=5,529$)

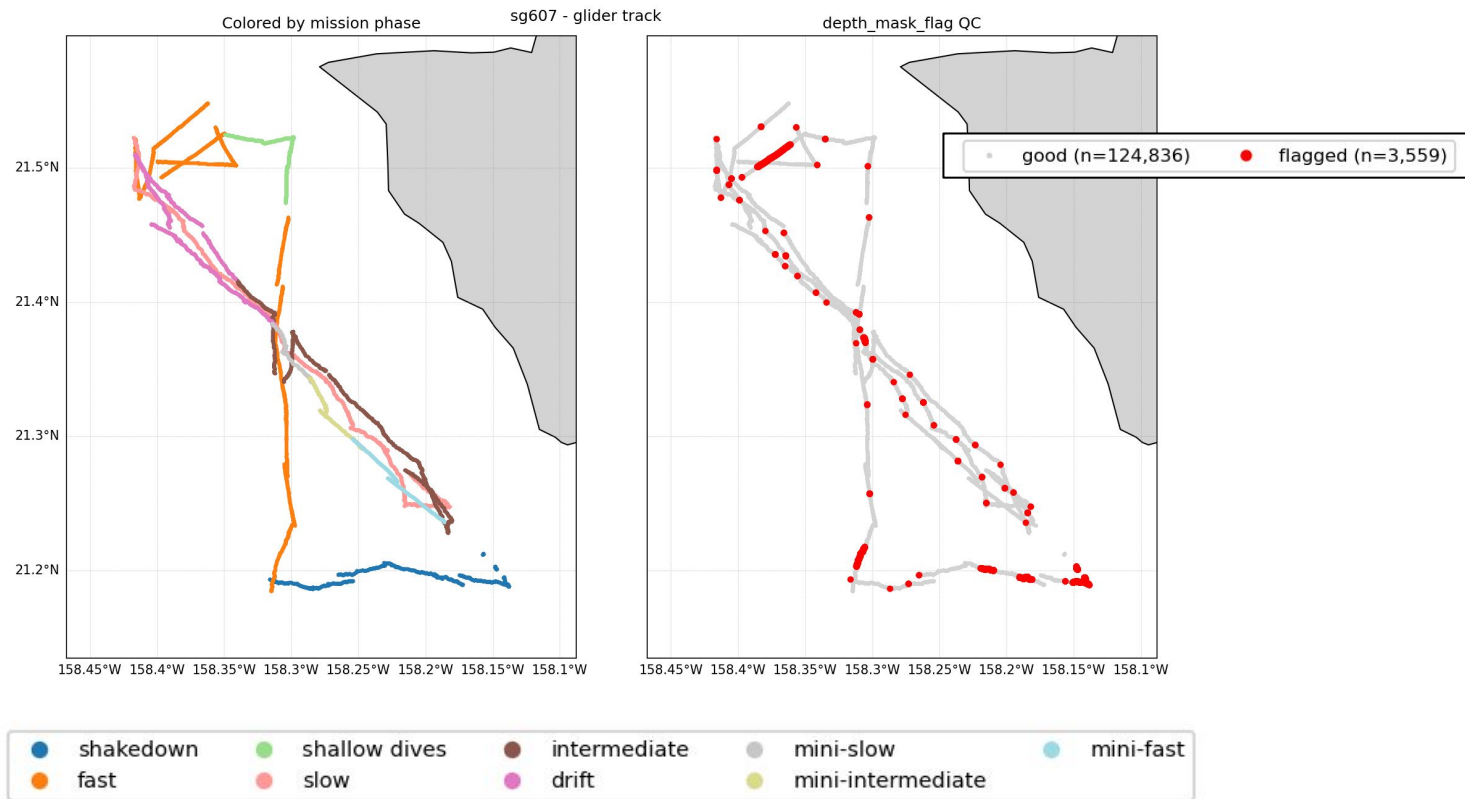
FULL-MISSION OVERVIEW

Depth over time by mission phase



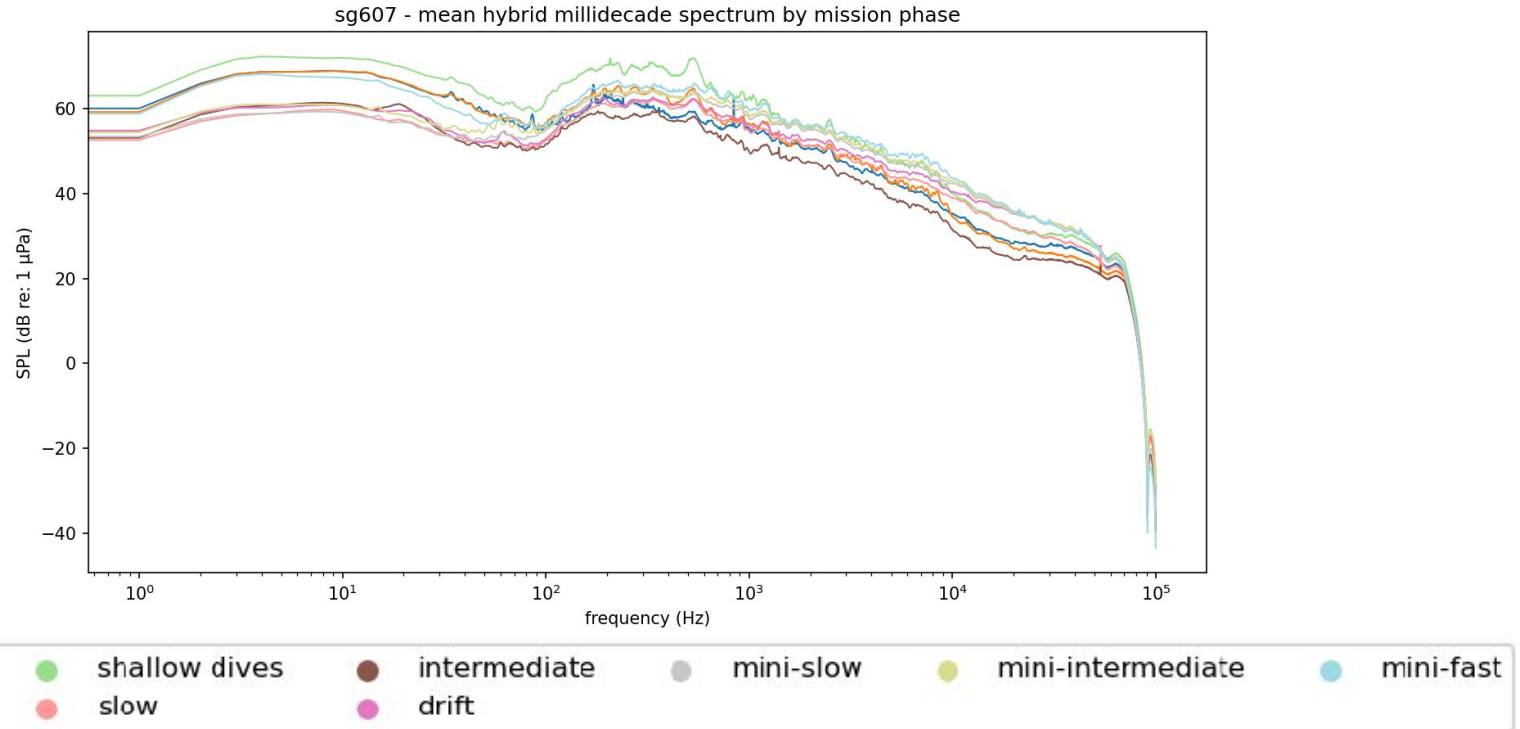
Full-deployment dive profile, colored by mission phase.

Glider track: phase & QC



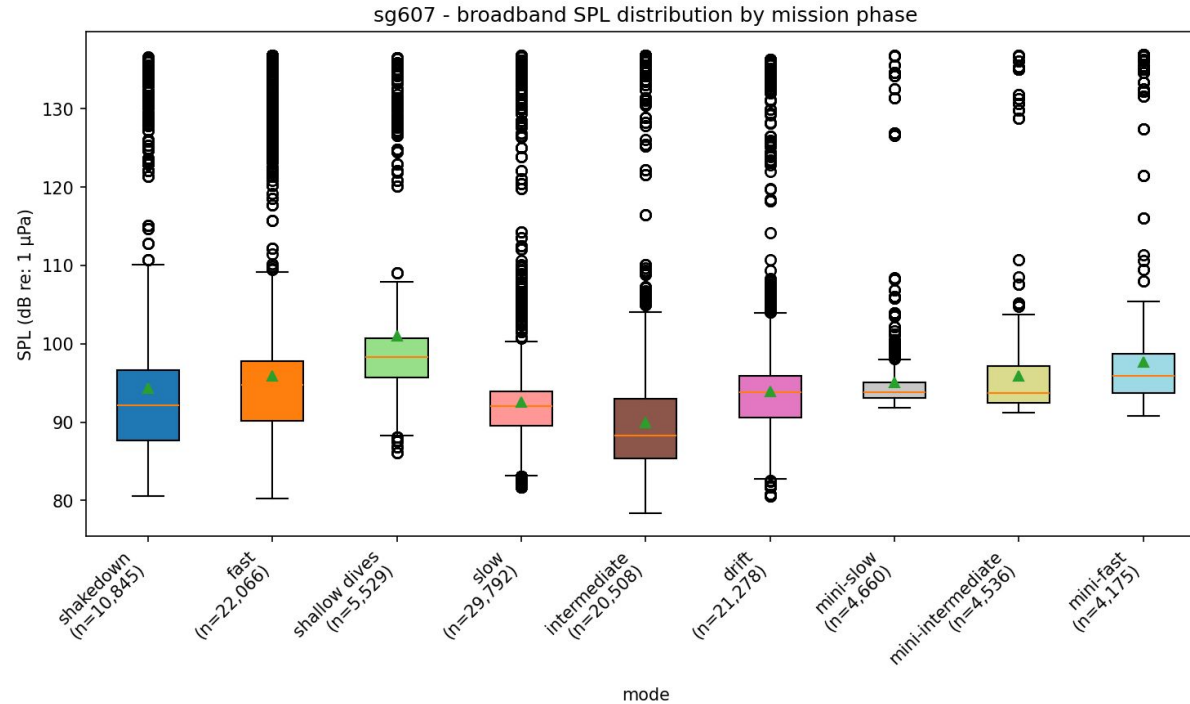
Full-deployment glider track (longitude vs latitude): left colored by mission phase, right highlighting rows flagged by the depth_mask_flag QC check.

Mean spectrum by mission phase



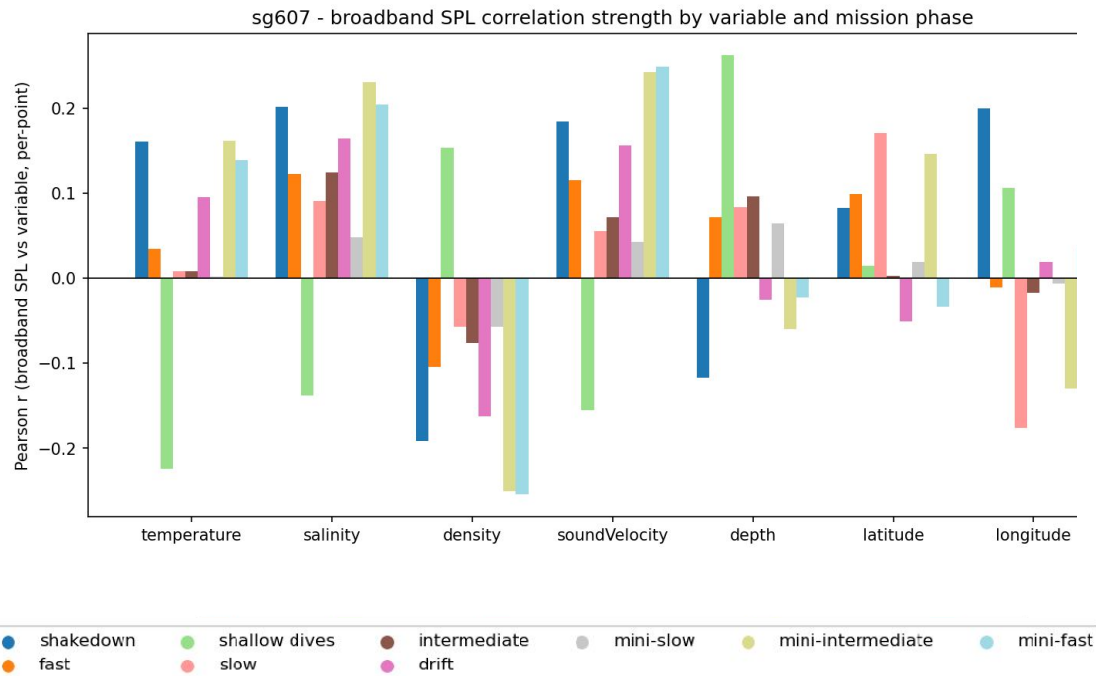
Mean hybrid millidecade noise spectrum for each mission phase - frequency (log) vs SPL, one line per mode.

Broadband SPL by mission phase



Distribution of broadband SPL for each mission phase.

Broadband SPL Correlation Strength by Variable and Mode

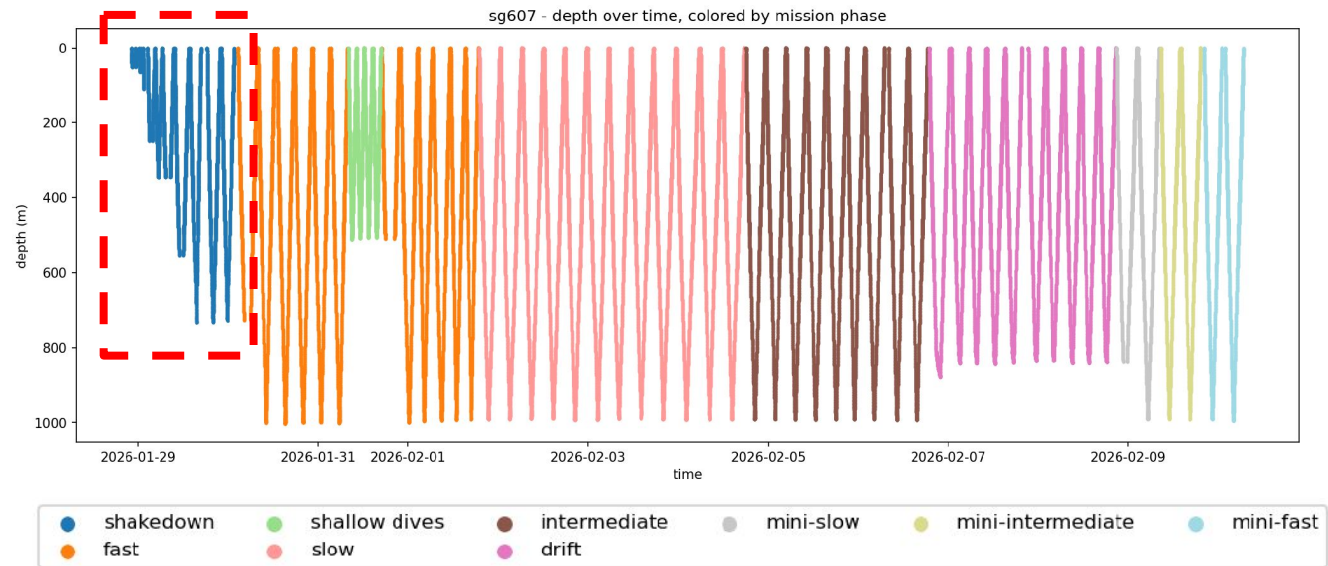


Real per-point Pearson r between broadband SPL and each raw variable, grouped by mission phase.

Key Takeaways & Applications

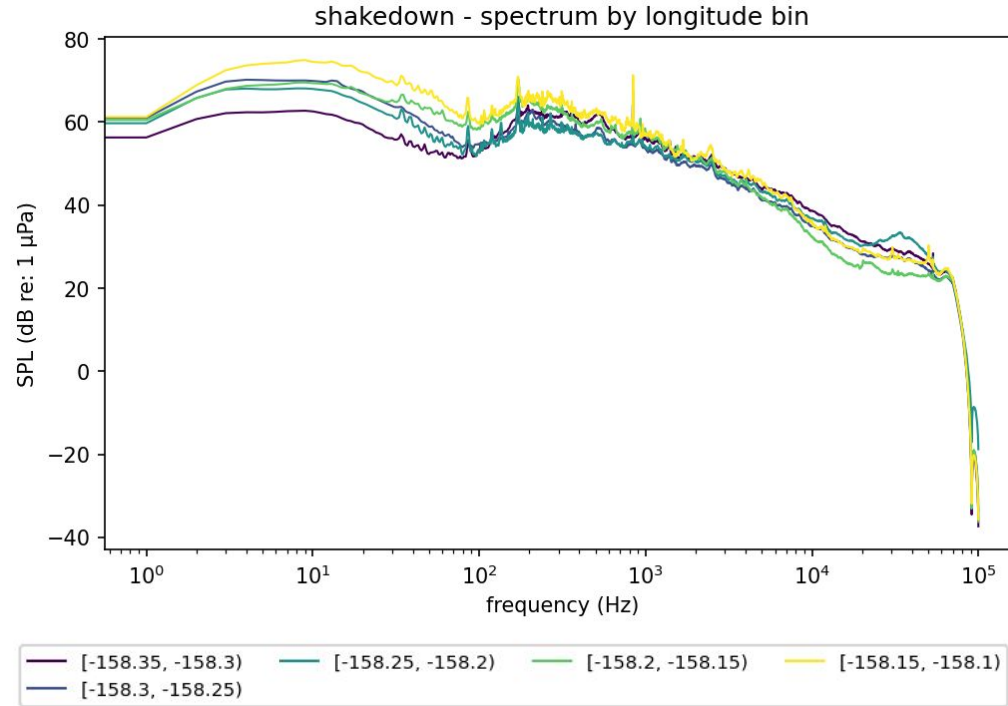
Shows the sign and magnitude of correlation between glider variables and broadband SPL across modes.

- **Regional Mission Planning:** Helps guide location-specific decisions (e.g., shallow dives in HI vs. deep dives off NJ).
- **Mode Optimization:** Highlights key variable influences per water mass to refine mission setup.
- **Platform Capabilities:** Evaluates dive/climb/hover impacts for gliders with depth-hold features.
- **Statistical Modeling:** Informs variable selection for Python OLS regression models.



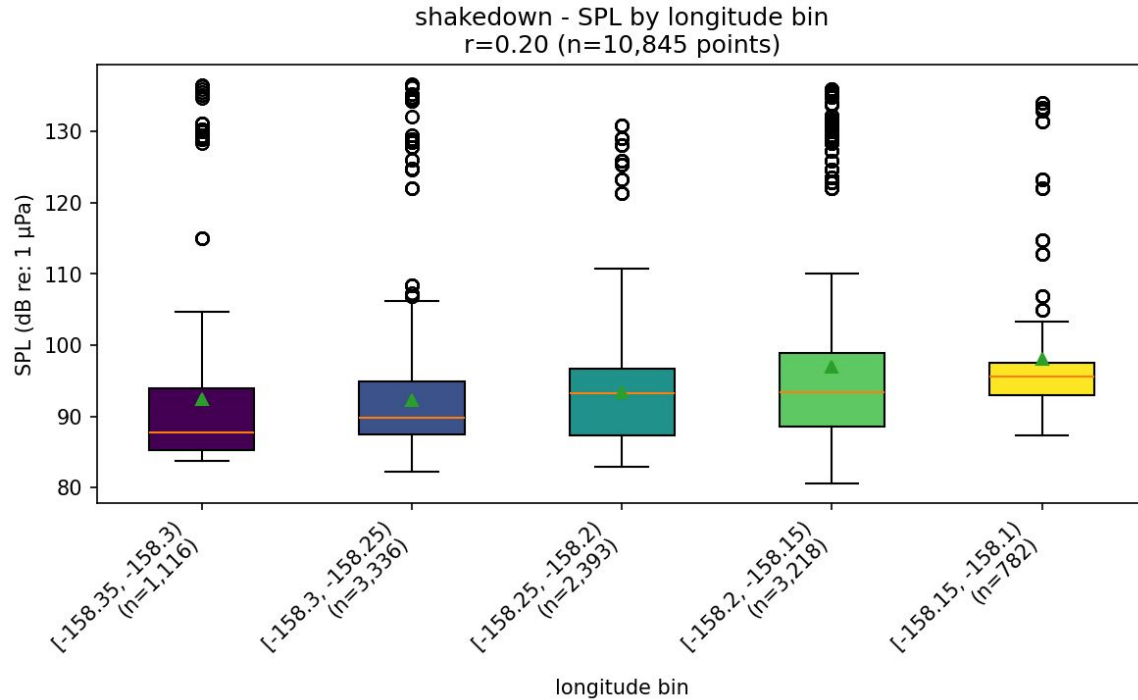
SHAKEDOWN DIVES

shakedown - spectrum by longitude bin



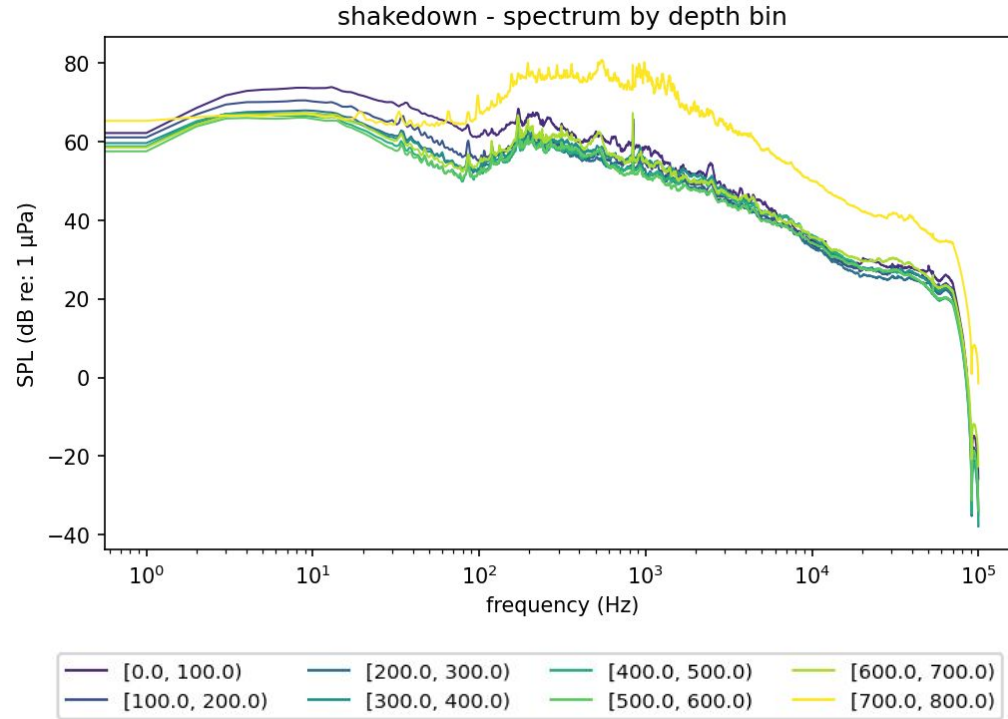
Mean noise spectrum during "shakedown" (frequency, log scale, vs SPL), one line per longitude bin.

shakedown - SPL by longitude bin



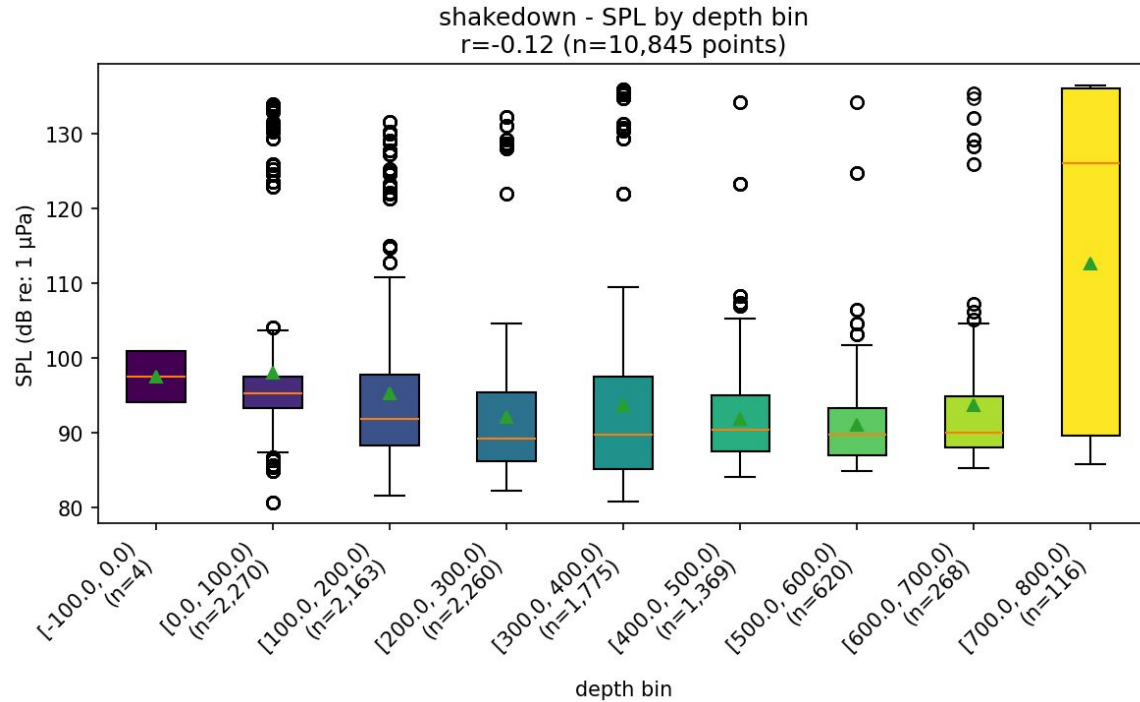
Broadband SPL distribution during "shakedown", split by longitude bin. Title shows the real per-point Pearson r between raw longitude and raw SPL for this mode.

shakedown - spectrum by depth bin



Mean noise spectrum during "shakedown" (frequency, log scale, vs SPL), one line per depth bin.

shakedown - SPL by depth bin



Broadband SPL distribution during "shakedown", split by depth bin. Title shows the real per-point Pearson r between raw depth and raw SPL for this mode.

Lessons Learned

Quality assessment of instrument noise proved tractable by examining audio characteristics against instrument and environmental features, offering a rapid way to flag regions of poor data quality. Leveraging cloud-based access to data through JupyterHub-formatted code further streamlined this critical first QA/QC step.

Next Steps

- Download filter: Exclude periods with elevated SPL when downloading data
- Finer diagnostic resolution
 - Third-octave bands
 - Bands linked to specific sources (e.g., flow noise)
- Calculate changes in noise level in species-specific bands to estimate likely changes in detection range
- Incorporate ship noise (AIS)

Future work will be able to determine whether these conditions significantly reduce the probability of signal detection.

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References

We used AI for coding assistance!

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